

Part 8: Regression Topics

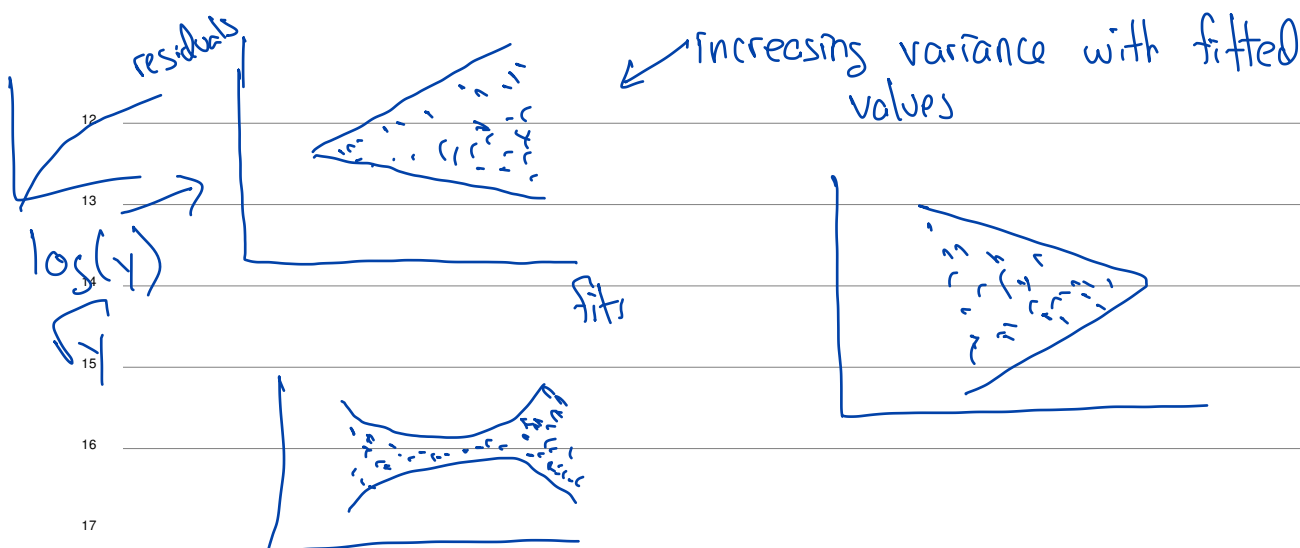
Heteroskedasticity

To this point, an assumption of our model has been that all of the model errors have the same variance; this condition is called **homoskedasticity**.

The opposite case, the presence of **heteroskedasticity** is a serious concern. The validity of the confidence intervals, hypothesis tests, and so forth rests on this assumption.

The plot of residuals versus fitted values is a useful tool for diagnosing heteroskedasticity.

Exercise: What shapes would one expect to see in the plot of residuals versus fitted values when heteroskedasticity is present?



- 1 There are two major strategies for dealing with heteroskedasticity.
- 2 First, the response variable can be **transformed**, meaning you would use
- 3 $g(y)$ instead of y as the response variable. The two most popular choices
- 4 for $g()$ are the logarithm, and the square root.
- 5 There is an accompanying Jupyter notebook that shows a simple example
- 6 in which the log transformation helps with heteroskedasticity.

The other major strategy for dealing with heteroskedasticity is to use **weighted least squares**. Briefly stated, instead of minimizing the residual sum of squares, one minimizes

$$\text{WRSS}(\beta) = \sum_{i=1}^n w_i (y_i - \hat{y}_i)^2$$

where the weights are chosen to deemphasize those observations for which the variance is larger.

In fact, the optimal choice is $w_i \propto 1/\sigma_i^2$, where σ_i^2 is the variance for the i^{th} model error.

$$V(\epsilon_i)$$

One would not typically know the σ_i^2 , but it is sometimes the case that you are willing to assume that $\sigma_i^2 = \sigma^2/m_i$. Then, the weight can be chosen as $w_i = m_i$. Or, the σ_i^2 could be estimated from the residuals.

Exercise: Explain why it may be the case that you would be willing to assume that $\sigma_i^2 = \sigma^2/m_i$.

If Y_i is a sample mean of m_i observations.

$$V(Y_i) = \frac{\sigma^2}{m_i} \quad (\text{Recall: } V(\bar{X}) = \frac{V(X_i)}{n})$$

$$\text{Use } w_i = \frac{1}{V(Y_i)} = m_i/\sigma^2 \propto m_i$$

Influential Observations

One source of concern is the possibility that one (or a few) observations are unduly **influential** on the resulting model fit. Of course, we expect that each data point should have some effect on the resulting fit. But no one observation should be overly influential; we'd prefer that each contribute a roughly equal amount to determining the final estimates.

A natural way of quantifying the influence of observation j is to consider the following question:

By how much do the fitted values $\hat{Y}_1, \hat{Y}_2, \dots, \hat{Y}_n$ change if observation j is excluded from the training set, i.e., it is not used in fitting the model?

This question is answered by **Cook's Distance**.

First, let $\hat{Y}_{i(-j)}$ denote the fitted value for observation i when using the model that **excludes** observation j from the fitting. Cook's Distance for observation j is then

$$D_j = \sum_{i=1}^n (\hat{Y}_i - \hat{Y}_{i(-j)})^2 / (p+1)\hat{\sigma}^2$$

Handwritten notes:
 - Arrow from $\hat{Y}_{i(-j)}$ to $\hat{Y}_{i(-j)}$: model with obs. j
 - Arrow from $p+1$ to $p+1$: # of predictors
 - $p+1 = \# \text{ of cols. in } \tilde{X}$

In Python, we can find Cook's Distance using

```
holdlm.get_influence().cooks_distance[0]
```

where `holdlm` is the output from the `statsmodels` function used previously.

Fortunately, calculating Cook's Distance for an observation does not actually require refitting the model with that observation removed. It is possible to show that ~~(in the case of simple linear regression),~~

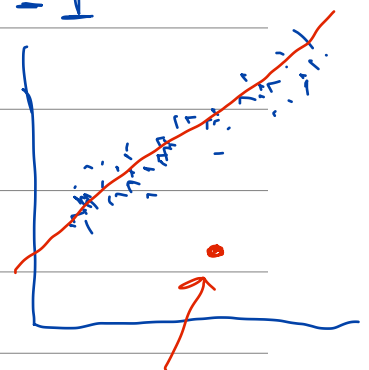
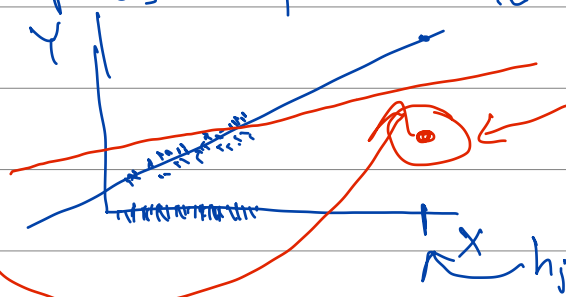
$$D_j = \frac{\hat{\epsilon}_j^2}{(p+1)\hat{\sigma}^2} \left[\frac{h_{jj}}{(1-h_{jj})^2} \right]$$

where h_{jj} is the previously-defined j^{th} leverage.

large if response value is "out of line"
potential to be influential
large if h_{jj} is large

Exercise: Consider the given expression for D_j . What does it say about what it takes to make an observation influential?

h_{jj} is large when an obs. has a great deal of potential to be influential, depends only on X , $\frac{1}{n} \leq h_{jj} \leq 1$



Handling Influential Observations

Influential observations are almost always an **outlier** in some sense; this is what causes them to have more influence than other observations.

First check to see if the influential point is not the result of a mistake. If so, the error should be corrected or, if that is not possible, the observation removed. But, **it is a mistake to arbitrarily remove an observation from a training set**. Instead, consider the following:

1. One should verify that there truly is a linear relationship between the response and predictor. A nonlinear model may be able fit more flexibly to the feature(s) that led to this outlier.
2. It may be the case that this observation is sufficiently different from the others that it would make sense to redefine the population of interest in such a way that this outlying (and other) cases are handled by a separate model. In other words, it may not be realistic to expect a single model to cover such a wide range of possibilities.
3. A transformation of the predictor and/or response can “pull in” an outlier so that it’s not so extreme.
4. An alternative approach to estimating the β_i may be preferable; for example, weighted least squares or robust regression.

Making Predictions

One of the fundamental motivations for using regression models is to make predictions for **future** observations of the response. In other words, consider a situation where you know the value of the predictor variable x , and seek to predict the response.

In the context of our simple linear regression model, if our value for the predictor is x^* , then we would start by finding

$$\hat{y}^* = \hat{\beta}_0 + \hat{\beta}_1 x^*$$

We have plugged in this new value for x into the line that has been fit. This is our best estimator for y^* , the true response for this object.

One should not be satisfied with only this, however: We should quantify the amount of error in this prediction.

In the case of simple linear regression, the **variance in the error in the prediction** is

$$V(\hat{y}^* - y^*) = \underbrace{\sigma^2}_{\textcircled{1}} + \underbrace{\sigma^2 \left[\frac{1}{n} + \frac{(x^* - \bar{x})^2}{(n-1)s_x^2} \right]}_{\textcircled{2}}$$

Exercise: Explain how the above decomposes the error in the prediction into two key sources.

① The irreducible error – the scatter around the regression line

② Error in $\hat{\beta}$'s – the fitted model is not equal to the true model

No matter how large n is source ① will remain.

Source ① dominates for most applications.

Hence the **standard error in the prediction** is

$$\cancel{SE(\hat{y}^*)} = \sqrt{\sigma^2 + \sigma^2 \left[\frac{1}{n} + \frac{(x^* - \bar{x})^2}{(n-1)s_x^2} \right]}$$

Of course, in practice σ^2 will be replaced with $\hat{\sigma}^2$.

$$V(\hat{\theta} - \theta_0) = V(\hat{\theta})$$

- 1 If the errors are assumed to be normally distributed, then a $100(1 - \alpha)\%$
2 **prediction interval for the future response** can be written as

3
$$\hat{y}^* \pm t_{\alpha/2, n-2} \text{SE}(\hat{y}^*)$$

- 4 Figure 1 shows a simple example of how the prediction interval will ap-
5 pear, as x^* ranges over the extent of the predictor space.

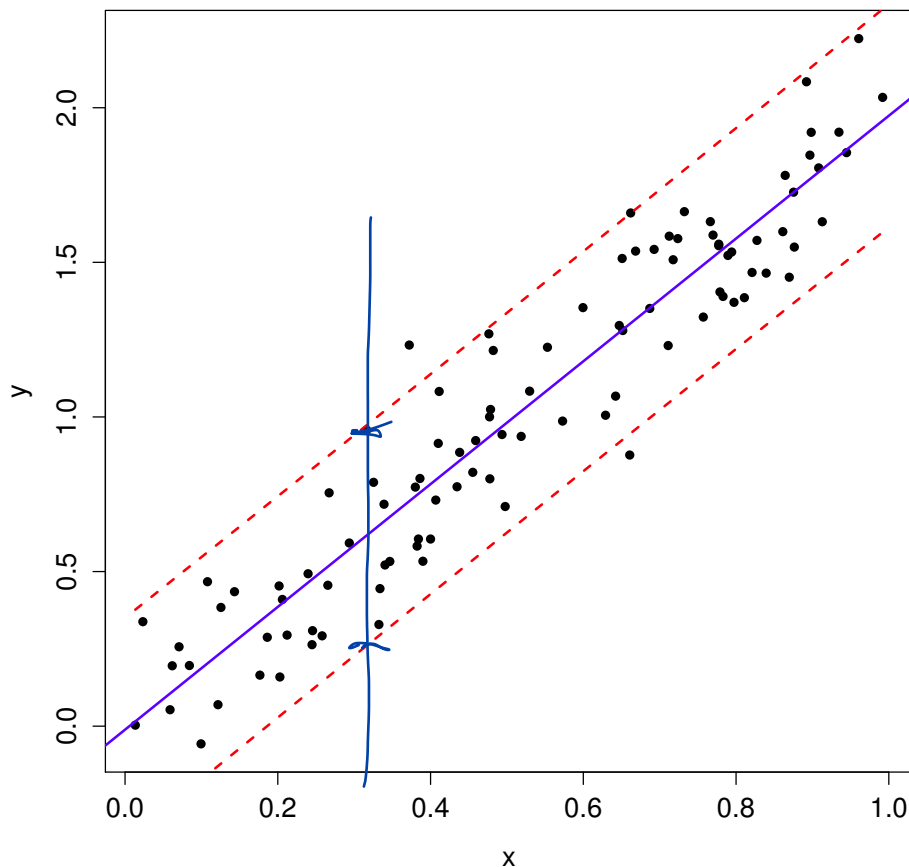


Figure 1: An example showing a regression line (solid line), along with the prediction bands (dashed lines).

Exercise: Discuss the role of the normality assumption in the validity of this interval. If the sample size is large, do you believe that the normality assumption can be relaxed?

If I assume $\epsilon_i \sim \text{Normal}(0, \sigma^2)$, this leads directly to the prediction interval

$$Y^* = \beta_0 + \beta_1 X^* + \epsilon$$

The scatter around the line has the normal dist, variability driven by ϵ

No CLT will help when ϵ are not normal

If using this theory for PIs, make sure normality assumption is valid